

AI Surveillance System Documentation – Staff Identification with Name Tag Detection

1. Executive Summary

This document details an advanced AI surveillance system engineered for **zero-shot staff identification**. The system is designed for real-world conditions, capable of identifying personnel using a single sample image of a name tag without any model retraining.

The core of the solution is an intelligent three-stage pipeline that addresses critical operational challenges:

- **Adaptive Performance:** It dynamically analyzes video quality to apply enhancements only when necessary, balancing processing speed with accuracy.
- **Multi-Modal Reasoning:** It fuses evidence from both text-based descriptions and direct visual matching, ensuring high-confidence classifications.
- **Explainable AI (XAI):** It provides clear visual evidence for its decisions by highlighting the exact region on a person that matches the name tag template, moving beyond "black box" predictions [1].

The system requires no fine-tuning to adapt to new name tags and delivers a comprehensive set of outputs, including an annotated video and a folder of automatically generated staff thumbnails for review.

Tech Stack: PyTorch, Ultralytics YOLOv8, Hugging Face Transformers (CLIP), OpenCV

2. System Architecture & Workflow

Stage 1: Smart Frame Enhancement

Low-quality CCTV frames are a major challenge. Instead of blindly applying enhancement filters to all frames, this system analyzes frame clarity using Variance of **Laplacian** [2].

- Frames with sharpness **below a threshold** are enhanced using:

Denoising → CLAHE → Sharpening

- High-quality frames bypass enhancement to optimize speed.

Result: Clean frames are processed faster; blurry ones are intelligently restored.

Stage 2: Person Detection using YOLOv8

- Model: yolo11n.pt
- Detection Target: People in video frames
- Confidence Threshold: Tuned to reduce false positives
- Output: The pixel coordinates for each detected person are passed to the next stage for identification.

Stage 3: Multi-Modal Staff Identification (Zero-Shot Learning)

This is the core of the zero-shot learning capability. For each detected person, CLIP generates two distinct forms of evidence to make a final decision.

A. Visual Evidence (Patch-Level Template Matching)

- **Template Encoding:** On startup, a single reference image (nametag.jpg) is encoded into a feature vector using CLIP's image encoder.
- **Patch Decomposition:** The cropped image of the detected person is conceptually divided into a 7x7 grid of patches.
- **Similarity Search:** CLIP calculates the cosine similarity between the reference name tag vector and the feature vector of each of the 49 patches.
- **Visual Score:** The highest similarity score among all patches is retained as the `image_score`

B. Textual Evidence (Semantic Description Matching)

This method provides contextual understanding.

1. **Prompting:** The cropped person image is compared against two semantic text prompts:
 - **Positive Prompt:** "a person with a small white rectangle on their chest"
 - **Negative Prompt:** "a person with no badge or tag on their clothes"
2. **Semantic Score:** CLIP determines the similarity of the image to each prompt, and the difference between these scores is used as the `text_score`.

Final Decision:

A weighted fusion of the two evidence scores produces the final confidence value.

Total Score = $(0.7 * \text{image_score}) + (0.3 * \text{text_score})$

If this Total Score surpasses the defined `STAFF_CONFIDENCE_THRESHOLD`, the individual is classified as "Staff".

3. Key Innovation: Multi-Modal Evidence with Visual Search

- **True Zero-Shot Capability:** The system requires **no re-training or fine-tuning**. To identify a different name tag, one only needs to replace the `nametag.jpg` file. This is made possible by the generalized world knowledge embedded within the CLIP model.
- **Explainable AI (XAI) for User Trust:** The system does not just provide a label; it provides proof. By drawing a yellow rectangle on the **exact patch** that produced the highest visual similarity score, it shows the user *why* it decided, making the results transparent and verifiable.
- **Configurable & Adaptable:** Key parameters (THRESHOLDS, WEIGHTS, PATHS) are centralized at the top of the script, allowing for easy tuning and adaptation to different operational environments without altering the core logic.
- **Automated Evidence Collection:** The system automatically saves a cropped thumbnail of every positively identified staff member to the `staff_thumbnails` directory, creating a log for security review.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning Transferable Visual Models From Natural Language Supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021, pp. 8748-8763.

- [2] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, 2005, vol. 2, pp. 60-65.